

Supplementary Material: Symbolic Candidate Relationship Specification

This supplementary material documents all 27 candidate process relationships screened across the six SWaT process stages (Stage 1: 5, Stage 2: 4, Stage 3: 6, Stage 4: 6, Stage 5: 5, Stage 6: 1; total 27), the audit criterion applied to each, and the resulting stage-level and cross-stage decisions. Of the 27 candidates, 16 were retained at the stage level, yielding six CORE features and ten RETAIN_WITH_CAVEAT features after the cross-stage audit. Two candidates were classified LIMITED and nine were rejected.

Symbolic Scoring Methodology

Empirical two-sided CDF score. Continuous production symbolic features, except the binary actuator-mismatch feature S2-R3B, use the empirical score

$$\text{score}(x) = 1 - \min(2 \min(F(x), 1 - F(x)), 1), \quad F(x) = \frac{\text{rank}(x \mid \mathcal{R})}{|\mathcal{R}|}, \quad (1)$$

where $\mathcal{R}$ is an empirical reference sample constructed exclusively from TRAIN-NORMAL observations. The score is small for values typical of the normal reference and approaches one toward either tail.

Reference distributions are conditioned according to the process relationship. State-conditioned references use target-sensor values under steady actuator states; transitional states of three-state valves are excluded from steady-state scoring. Slope-conditioned references use rolling differences $x_t - x_{t-w}$, either conditioned on actuator state or evaluated unconditionally for mass-balance relations. Sensor-sensor relations use ratios or differences as residual quantities. S2-R3B is not CDF-based and instead uses the binary mismatch

$$s_t = 1[P205_t \neq P203_t]. \quad (2)$$

The fixed diagnostic search grids are

$$w \in \{10, 30, 60, 120, 300\} \text{ s}$$

for rolling slopes,

$$\{0, 1, 2, 5, 10, 15, 20, 30, 45, 60, 90, 120\} \text{ s}$$

for event-response offsets, and

$$\{-30, -15, -10, -5, 0, 5, 10, 15, 30\} \text{ s}$$

for sensor-residual lag analysis. Event-response offsets provide diagnostic evidence rather than production scores. Likewise, a discovery-stage best lag is diagnostic only; retained residual features are scored at lag zero.

All normal-reference distributions are estimated from TRAIN-NORMAL observations only. Rolling and slope calculations are performed independently within chronological partitions so that no calculation crosses a partition boundary.

Stage-level screening and final CORE admission are separate decisions. Stage-level screening assigns each candidate to RETAIN, LIMITED, or REJECT using relation-specific support criteria and coverage checks. The subsequent cross-stage audit applies additional checks for reference sufficiency, temporal coverage, saturation, lag-boundary effects, redundancy, distribution shift, and explanatory strength. A retained feature with no identified cross-stage caveat is assigned to CORE; otherwise it is retained with an explicit caveat.

Candidate Identities and Decisions

Table 1: All 27 candidate process relationships. AS = actuator-sensor; DP = dynamic process; AC = actuator coupling; SS = sensor-sensor. RET/LIM/REJ denote stage-level RETAIN/LIMITED/REJECT; CORE/RWC denote final CORE/RETAIN.WITH.CAVEAT status.

Rule	Stage	Variables	Cat.	Stage	Final	Caveat / rejection reason
S1-R1	1	MV101, FIT101	AS	RET	CORE	—
S1-R2	1	P101, LIT101	DP	RET	CORE	—
S1-R3	1	P102, LIT101	DP	RET	RWC	Thin TRAIN-NORMAL reference ($n = 45 < 100$)
S1-R4	1	FIT101, LIT101	DP	RET	CORE	—
S1-R5	1	P101, P102	AC	REJ	—	Coupling rate = $0.00 < 0.8$
S2-R1	2	MV201, FIT201	AS	RET	CORE	—
S2-R2	2	P205, AIT203	AS	LIM	—	Fallback score saturated in VALIDATION (mean $0.978 \geq 0.90$)
S2-R3B	2	P205, P203	AC	RET	CORE	—
S2-R4	2	P203, AIT202	AS	LIM	—	No fallback reaches 90% VALIDATION coverage (best 0.03%)
S3-R1	3	P302, FIT301	AS	RET	RWC	Distribution shift ($\Delta = +0.351$)
S3-R2	3	P302, DPIT301	AS	RET	RWC	Distribution shift ($\Delta = +0.464$)
S3-R3	3	FIT301, DPIT301	SS	REJ	—	$ \rho = 0.196 < 0.3$
S3-R4	3	FIT301, LIT301	DP	RET	RWC	Low explained variance ($7.75\% < 15\%$)
S3-R5	3	P302, LIT301	DP	REJ	—	Overlap = $0.707 \geq 0.5$
S3-R6	3	MV304, P302	AC	REJ	—	Coupling rate = $0.064 < 0.8$
S4-R1	4	UV401, AIT402	AS	REJ	—	Overlap = $0.400 \geq 0.25$
S4-R2	4	P402, FIT401	AS	RET	RWC	Thin TRAIN-NORMAL reference ($n = 65 < 100$)
S4-R3	4	P402, LIT401	DP	REJ	—	$ \text{SMD} = 0.037 \leq 0.5$
S4-R4	4	FIT401, LIT401	DP	REJ	—	Explained variance ≈ 0
S4-R5	4	P402, P403	AC	REJ	—	Only $n = 3$ transitions (< 10)
S4-R6	4	AIT401, AIT402	SS	REJ	—	$ \rho = 0.030 < 0.3$
S5-R1	5	P501, FIT504	AS	RET	RWC	Thin TRAIN-NORMAL reference ($n = 89 < 100$)
S5-R2	5	P501, PIT501	AS	RET	RWC	Thin TRAIN-NORMAL reference ($n = 89 < 100$)
S5-R3	5	FIT501, FIT503	SS	RET	CORE	—
S5-R4	5	PIT501, PIT503	SS	RET	RWC	Source sensors near-collinear ($\rho = 0.996 \geq 0.95$) and distribution shift ($\Delta = +0.494$)
S5-R5	5	AIT501, AIT504	SS	RET	RWC	Best diagnostic lag -30 s lies at the search-grid boundary
S6-R1	6	P602, FIT601	AS	RET	RWC	Minority actuator state represents only 0.85% of TRAIN

Mathematical Forms and Audit Evidence

Table 2: Symbolic form, TRAIN-NORMAL reference, and audit evidence for the 27 candidates. “Audit-only” denotes a rejected or limited candidate that did not become a production symbolic feature. Criteria A–G are defined below.

Rule	Mathematical form	TRAIN-N reference	Criterion and evidence
S1-R1	CDF, 3-state valve	off $n = 62,302$; on $n = 158,769$	A: SMD = 20.94, overlap = 0.00
S1-R2	CDF, pump slope, $w = 120$ s	off $n = 55,825$; on $n = 166,398$	B: SMD = -1.78 , overlap = 0.07
S1-R3	CDF, pump slope, $w = 300$ s	off $n = 221,998$; on $n = 45$	B: SMD = -2.59 , overlap = 0.00
S1-R4	CDF, unconditional slope, $w = 60$ s	$n = 222,283$	C: $\rho = 0.449$, explained variance = 0.370
S1-R5	Audit-only coupling indicator	$n_{\text{trans-active}} = 125$	E: coupling rate = 0.00
S2-R1	CDF, 3-state valve	$n = 64$ normal transition sequences	F: consistency = 100.0%; VAL coverage = 99.58%; mean = $0.568 < 0.90$
S2-R2	CDF, 2-state fallback	$n = 64$ events	F: consistency = 98.4%; fallback mean = 0.978
S2-R3B	$1[P205 \neq P203]$	n/a	F: consistency = 100.0%; mismatch rate = 0.0094%
S2-R4	Audit-only event response	$n = 64$ events	F: consistency = 95.3%; best coverage = 0.03%
S3-R1	CDF, 2-state actuator	$n = 111$ events	A: SMD = 8.95, overlap = 0.00
S3-R2	CDF, 2-state actuator	$n = 111$ events	A: SMD = 7.49, overlap = 0.00
S3-R3	Audit-only residual	$n = 222,343$	D: CV = 0.392, $ \rho = 0.196$
S3-R4	CDF, unconditional slope, $w = 300$ s	$n = 222,043$	C: $\rho = -0.232$, explained variance = 0.078
S3-R5	Audit-only pump slope	$n = 35,399$	B: SMD = -0.85 , overlap = 0.707
S3-R6	Audit-only coupling indicator	$n_{\text{trans-active}} = 625$	E: coupling rate = 0.064
S4-R1	Audit-only state relation	$n = 1$ transition	A: SMD = -2.36 , overlap = 0.400
S4-R2	CDF, 2-state actuator	reference $n = 65$	A: SMD = 146.7, overlap = 0.00
S4-R3	Audit-only pump slope	$n = 65$	B: SMD = 0.037
S4-R4	Audit-only unconditional slope	$n = 222,043$	C: $\rho = -0.141$; explained variance $\approx 4.6 \times 10^{-5}$
S4-R5	Audit-only coupling indicator	$n_{\text{trans-active}} = 3$	E: $n < 10$
S4-R6	Audit-only residual	$n = 222,343$	D: CV = 0.023, $ \rho = 0.030$
S5-R1	CDF, 2-state actuator	reference $n = 89$	A: SMD = 5.82
S5-R2	CDF, 2-state actuator	reference $n = 89$	A: SMD = 3.48
S5-R3	CDF, residual ratio FIT501/FIT503, lag = 0	$n = 222,343$	D: CV = 0.016, $\rho = -0.828$
S5-R4	CDF, residual difference PIT501–PIT503, lag = 0	$n = 222,343$	D: CV = 0.018, $\rho = 0.996$
S5-R5	CDF, residual ratio AIT501/AIT504, lag = 0	$n = 222,343$	D: CV = 0.103, $\rho = -0.758$; diagnostic best lag = -30 s
S6-R1	CDF, 2-state actuator	$n = 64$ events	A: SMD = 3.10, overlap = 0.00

Audit Criteria

No universal numerical threshold is applied to all candidate relationships. The audit uses relation-specific criteria:

- **Criterion A (actuator–sensor state support):** $|\text{SMD}| > 1.0$, IQR overlap < 0.25 , and reference $n \geq 10$.
- **Criterion B (pump–slope support):** $|\text{SMD}| > 0.5$, overlap < 0.5 , and $n \geq 10$.
- **Criterion C (mass-balance/dynamic support):** $|\rho| > 0.2$, $p < 0.01$, explained variance > 0.04 , and $n \geq 1000$.
- **Criterion D (sensor–sensor consistency):** stability CV < 0.5 , $|\rho| > 0.3$, and $p < 0.01$.
- **Criterion E (actuator coupling):** coupling rate ≥ 0.8 with at least ten supported transitions.
- **Criterion F (Stage-2 event/consistency audit):** at least ten supported events and expected-direction consistency $\geq 95\%$. RETAIN additionally requires a continuously evaluable feature with VALIDATION coverage $\geq 90\%$ and VALIDATION-NORMAL mean score < 0.90 ; otherwise the relation is classified LIMITED.
- **Criterion G (cross-stage CORE audit):** the 16 stage-level retained features are checked for thin normal references ($n < 100$), actuator minority-state support below 5% of TRAIN, explained variance below 15%, a discovered lag at the ± 30 s search boundary, cross-feature score redundancy ($|\rho| \geq 0.90$), source-sensor redundancy ($|\rho| \geq 0.95$), TRAIN-to-VALIDATION distribution shift ($|\Delta| \geq 0.30$ or VALIDATION median ≥ 0.85), coverage below 90%, and saturation (VALIDATION-NORMAL mean ≥ 0.90). Features with none of these caveats form CORE; otherwise retained features are documented as RETAIN_WITH_CAVEAT when the identified issue does not require exclusion.

For Criteria A–E, candidates that pass the relation-specific support test are additionally required to achieve at least 90% VALIDATION coverage with a VALIDATION-NORMAL mean symbolic score below 0.90. A failure of this coverage/saturation gate yields LIMITED rather than RETAIN. No Criteria A–E candidate that otherwise passed its support test triggered this gate in the present study.

Reconciliation

The complete audit reconciles as

$$27 = 16 \text{ RETAIN} + 2 \text{ LIMITED} + 9 \text{ REJECT},$$

with the 16 retained relationships further divided into

$$16 = 6 \text{ CORE} + 10 \text{ RETAIN_WITH_CAVEAT}.$$

The six CORE relations are S1-R1, S1-R2, S1-R4, S2-R1, S2-R3B, and S5-R3.